

Social Media Data Ingestion

Data Sources

Live APIs (pulled every run)

Arctic Shift - *Reddit posts + comments. Free. Live and historical.*
FetchLayer - *Reddit + X/Twitter. Paid, ≈USD 15 per 7 000 pulls.*
StockTwits - *public stream. Free, no key*

Historical / back-fill (loaded once)

Reddit comment archive dumps + academic and Hugging Face tweet corpora. **From 2017-2026.** (some time frames, e.g pre 2023 before Reddit official API, have much more data)

BRONZE Raw Data

API Download:

One line of **JSON per post**, whole file **zstd-compressed**. Nothing is edited, renamed or dropped at this stage.

data/raw/RedditLive/reddit_live_2026-07-19.jsonl.zst
data/raw/RedditComments/comments_2026-07-24.jsonl.zst

Text compresses ≈5×6 months of Reddit comments is 43 MB
Every source keeps a list of ids it has already seen plus a per-subreddit watermark, so a re-run adds only what is new nothing is downloaded or counted twice.

Reddit:

```
{"id":"1v0dapl", "created_utc":1784424655}
{"author":"Wrench365", "subreddit":"investing"}
{"title":"Index Mutual Funds E Trade", "selftext":"..."}
{"score":1, "num_comments":1, "upvote_ratio":1.0}
{"permalink":"/r/investing/...", "_backend":"official"}
+ 100 further Reddit fields
```

Stock Twits

```
"id":660079305, "body":"ENTERED @ 0.81 $AAPL"
{"entities":{"sentiment":{"basic":"Bullish"}}}
id,created_at,text,author,likes
207653955377721802,Mon Jul 13 05:29:44 +0000 2026,
"In 1998, Warren Buffett..." ,Alokkumarzz,0
```

SILVER Conformed Data

data/processed/posts.parquet

- **Conform** - all four platforms rewritten into the same 9 columns, so Reddit, StockTwits and X can be counted together at all
- **Namespace ids** - st_, x_ prefixes, so two platforms can never collide; dedupe on id, first seen wins.
- **One clock** - every timestamp becomes a UTC calendar day, whatever format it arrived in
- **Drop** empty and deleted posts (deleted before the scrape)
- **Extract tickers** \$GME always; bare GME only if it is a real listed symbol and not an English word (EDGE, LOAN, MOON blocked). *I used wordfreq for this (comparing a ticker vs the frequency it is used in natural language AND capitalisation rate)*
- **Score sentiment** -1 to +1 per post using finVADER, finance + WallStreetBets word list, cached by id so a post is never re-scored
- **Tag** themes 39 of them (semiconductors, uranium_nuclear, ai, ...) using simple regex.

Main Parquet

```
id | date | author | score | subreddit | title | selftext | num_comments | source
1v0dapl | 2026-07-19 | Wrench365 | 1 | investing | Index Mutual Funds E Trade |
... | 1 | reddit
st_660079305 | 2026-07-27 | trader99 | 0 | stocktwits | ENTERED @ 0.81 $AAPL
| | 0 | stocktwits
x_2076539553 | 2026-07-13 | Alokkumarzz | 0 | x_twitter | In 1998, Warren
Buffett... | | 0 | x
```

Per-post sentiment cache

```
sentiment_scores_finvader_wsb.parquet
id | sentiment
1v0bhlw | 0.00
1v0dapl | 0.41
```

GOLD Extractable data

One row per day topics and sentiment

- *daily_ticker_counts* - how many posts mentioned each ticker?
- *daily_ticker_sentiment* - how bullish they were?
- *daily_theme_counts* / *daily_theme_sentiment* - the same two questions per theme (from FIMA available ETF list)
- *daily_term_counts* - trending words and phrases, rolling 365 days

This is the only layer that leaves the data ingestion pipeline. It carries no post text, no authors and no post ids which is what keeps the shared folder under 25 MB instead of several GB, and lets the whole upstream research and dashboard stack run with no access to the raw posts at all. *2017 → today is ≈300k rows*

daily_ticker_counts.parquet

```
date | ticker | mention_count
2026-07-27 | NVDA | 412
```

daily_ticker_sentiment.parquet

```
date | ticker | n_posts | avg_sentiment | net_bullish
2026-07-27 | NVDA | 412 | 0.21 | 0.38
```

daily_theme_counts.parquet

```
date | theme | mention_count
2026-07-27 | semiconductors | 138
```

daily_term_counts.parquet

```
date | term | mention_count
2026-07-27 | harmonic drive | 9
```